

Gowtham Sai Chimmana

AI/ML Engineer | Agentic AI & Generative AI | LLMs, RAG

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SUMMARY

AI/ML Engineer specializing in Agentic AI and Generative AI — built a multi-step, self-critiquing LangGraph agent achieving a 100% eval pass rate, an open-source MLflow observability/eval library adopted across two production agents, plus production RAG and generative LLM systems. Full-stack range from model experimentation to production deployment. MS Computer Science candidate at the University of South Florida (May 2026), authorized to work in the United States (OPT).

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, BERT, RoBERTa, NLP, LLMs, RAG, MLflow

LLM & Agents: LangChain, LangGraph, LlamaIndex, Pinecone, ChromaDB, Prompt Engineering, OpenAI API, Groq, Ollama, E2B

Languages: Python, JavaScript, TypeScript, SQL, Bash, Java, C++

Backend & MLOps: FastAPI, Docker, Git, CI/CD, Linux, PostgreSQL, MongoDB, Microsoft Azure, PySpark

Frontend: React.js, Next.js, REST APIs, Responsive Design, Figma

PROJECTS

AgentFlow Studio — Autonomous AI Agent System with Human-in-the-Loop Control — **Python, FastAPI, LangGraph, PostgreSQL, E2B, Next.js, Docker**

- Architected a self-correcting 5-node LangGraph state machine (plan → execute → sandbox → evaluate → report) that diagnoses failed steps and retries under a bounded error budget, eliminating infinite-loop failure modes.
- Built human-in-the-loop approval checkpoints on LangGraph's interrupt() and PostgreSQL-backed checkpointing, pausing execution only for critical/irreversible actions and durably resuming from that exact step even after a process restart.
- Isolated all LLM-generated code execution in per-thread E2B sandboxes with enforced timeouts and persistent session reuse across steps, removing host exposure to untrusted code; shipped end-to-end with FastAPI SSE streaming to a Next.js dashboard.
- Open-sourced [agent-observability-kit](#), an MLflow tracing + LLM-judge eval library, instrumenting this agent and AI Career Copilot.

AI Career Copilot — Agentic RAG System — **Python, LangGraph, FastAPI, FAISS, Groq, React, Adzuna API**

- Architected a multi-step LangGraph agent flow — retrieve live postings (Adzuna API) → match role via embedding similarity → ground analysis in retrieved job text (FAISS) → generate → self-critique → conditionally regenerate — delivering grounded, job-specific skill-gap analysis instead of a single-shot LLM response.
- Achieved a 100% eval pass rate and 1.0 average faithfulness score across 6 test scenarios by engineering an automated LLM-judge critique-and-regeneration loop as a conditional edge in the agent graph.
- Added MLflow tracing and an independent LLM-judge eval pass alongside in-graph self-critique, catching regressions via a second, unbiased signal.

AI/ML Visual Defect Inspection System — **Python, TensorFlow, Keras, FastAPI, TFLite**

- Achieved 96.4% classification accuracy across PASS/FAIL/REVIEW categories on a 2,000+ image manufacturing test set by training and deploying a CNN defect-detection model.
- Enabled real-time production inference by compressing the model to TFLite, cutting size 91.7% (127.87 MB to 10.66 MB) while retaining accuracy within 1.2% and achieving sub-100ms CPU latency via a FastAPI REST endpoint.

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant, Operating Systems — **University of South Florida**

Tampa, FL | Nov 2024 – May 2026

- Contributed to 2 active faculty publications in neural-symbolic AI, as measured by manuscript co-authorship credit, by supporting Prof. Valentina Korzhova's research in text parsing, semantic reasoning, and linguistic semantics.
- Achieved 92%+ course satisfaction ratings by instructing and mentoring 60+ graduate students per semester in Operating Systems topics including scheduling, memory management, concurrency, and synchronization.
- Cut grading turnaround by 35% by designing and building Python grading automation for OS simulation projects, while resolving 100+ student queries per semester through office hours.
- Strengthened student mastery of core OS concepts by designing 12+ coding assignments and labs aligned with ACM CS curriculum standards.

Frontend Engineer — **Phoenix Global**

Remote | Jun 2022 – May 2024

- Shipped the company's core tutoring platform UI/UX from the ground up, serving thousands of daily active students and tutors, as the team's sole frontend engineer using JavaScript, HTML, and CSS.
- Gave staff live operational visibility into tutor availability, course data, and student progress by building real-time dashboards integrated with the backend LMS via REST APIs.
- Reduced manual QA time and data-entry errors by building Python/JavaScript automation for data validation and formatting between the LMS backend and dashboards.
- Increased user engagement and session duration by improving page load performance and mobile responsiveness across the platform.

EDUCATION

M.S. Computer Science — **University of South Florida** | Aug 2024 – May 2026 | GPA 3.50/4.0

Coursework: Machine Learning, NLP, Distributed Systems, Advanced Algorithms

B.Tech Computer Science & Engineering — **KL Deemed to be University, India** | Aug 2020 – May 2024 | GPA 8.57/10.0

CERTIFICATIONS

Microsoft Azure AI-900 | Microsoft Certified: Azure Fundamentals